

Brand and Region Graphs

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely $I(1)$ -like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

brand_io_rows=48

brand_econ_rows=1880

region_rows=171

Interpretation option: IO concentration, promo intensity, and regional dispersion.

Table: Brand_IO_Metrics

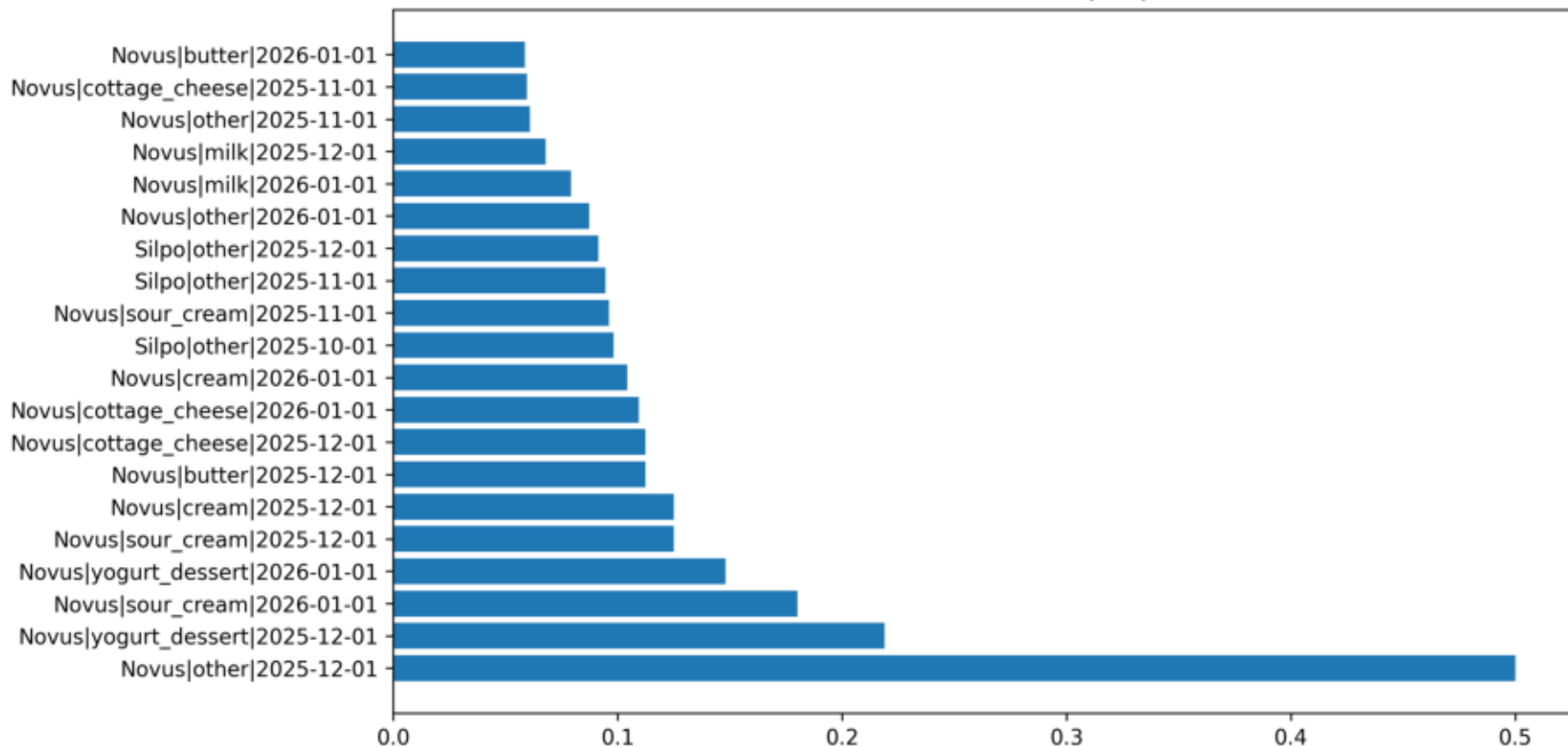
source	standardized_type	month	hhi_brand	top3_share	private_label_share	sku_count	n_obs
Novus	butter	2025-11-01	0.05601838552140192	0.3050847457627119	0.0	59	59
Novus	butter	2025-12-01	0.1124260355029586	0.38461538461538464	0.0	13	13
Novus	butter	2026-01-01	0.0586734693877551	0.24999999999999997	0.0	28	28
Novus	cottage_cheese	2025-11-01	0.05947872577411451	0.3283582089552239	0.014925373134328358	67	67
Novus	cottage_cheese	2025-12-01	0.11224489795918367	0.42857142857142855	0.0	14	14
Novus	cottage_cheese	2026-01-01	0.10941828254847642	0.4736842105263157	0.02631578947368421	38	38
Novus	cream	2025-11-01	0.024822180287414727	0.14457831325301204	0.012048192771084338	83	83
Novus	cream	2025-12-01	0.125	0.5	0.0	12	12
Novus	cream	2026-01-01	0.10416666666666663	0.45833333333333337	0.0	24	24
Novus	hard_cheese	2025-11-01	0.016879012345679012	0.13333333333333333	0.0022222222222222222	450	450
Novus	hard_cheese	2025-12-01	0.03790087463556851	0.20408163265306123	0.0	49	49
Novus	hard_cheese	2026-01-01	0.040583837664649326	0.25471698113207547	0.009433962264150943	105	106
Novus	milk	2025-11-01	0.0478522599734721	0.2727272727272727	0.0	99	99
Novus	milk	2025-12-01	0.06790123456790123	0.27777777777777778	0.0	18	18
Novus	milk	2026-01-01	0.07922136713444997	0.36170212765957444	0.0	47	47
Novus	other	2025-11-01	0.060815326353308094	0.3283582089552239	0.0	67	67
Novus	other	2025-12-01	0.5	1.0	0.0	2	2
Novus	other	2026-01-01	0.08723599632690543	0.393939393939394	0.0	33	33
Novus	sour_cream	2025-11-01	0.09607438016528924	0.4318181818181818	0.0	43	44
Novus	sour_cream	2025-12-01	0.125	0.375	0.0	8	8
Novus	sour_cream	2026-01-01	0.18000000000000002	0.65	0.0	20	20
Novus	yogurt_dessert	2025-11-01	0.05347439010290208	0.3010752688172043	0.016129032258064516	186	186
Novus	yogurt_dessert	2025-12-01	0.21900826446280988	0.7727272727272727	0.0	22	22
Novus	yogurt_dessert	2026-01-01	0.14812611540749557	0.5853658536585366	0.0	41	41
Silpo	butter	2025-10-01	0.04457444547600139	0.27982162764771457	0.01839464882943144	170	1794
Silpo	butter	2025-11-01	0.043937430880431456	0.2739979445015416	0.018499486125385406	170	4865
Silpo	butter	2025-12-01	0.044002465675947885	0.2718360071301248	0.01871657754010695	162	1122
Silpo	cottage_cheese	2025-10-01	0.043909354958425476	0.25131348511383533	0.024080560420315235	218	2284
Silpo	cottage_cheese	2025-11-01	0.043919424761230244	0.24839927762272207	0.02462649811196848	220	6091
Silpo	cottage_cheese	2025-12-01	0.044933486920332937	0.2535919540229885	0.029454022988505746	205	1392
Silpo	cream	2025-10-01	0.041181992775132695	0.24230465666929757	0.008681925808997633	118	1267
Silpo	cream	2025-11-01	0.03885079598942614	0.2316690442253923	0.008559201141226819	128	3505
Silpo	cream	2025-12-01	0.03945560040049502	0.2344139650872818	0.008728179551122194	117	802
Silpo	hard_cheese	2025-10-01	0.051933216933845096	0.2910495814552479	0.06374758531873792	145	1553
Silpo	hard_cheese	2025-11-01	0.052037932551782407	0.29124618634123445	0.06313072048814833	148	4261
Silpo	hard_cheese	2025-12-01	0.052662662244763916	0.3007063572149344	0.06357214934409687	145	991
Silpo	milk	2025-10-01	0.03203793286785125	0.21061167747914736	0.012743280815569972	408	4316
Silpo	milk	2025-11-01	0.03329842826385409	0.21858294442087398	0.012728044123886296	415	11785
Silpo	milk	2025-12-01	0.033714085433954495	0.22108345534407028	0.01281112737920937	399	2732
Silpo	other	2025-10-01	0.09804676817006609	0.4928813559322034	0.022372881355932205	135	1475

Table: Prozorro_ByRegion

region	product	standardized_type	count	missing	imputed_share	mean	median	std	min	max	q05	q25	q75	q95	skew	kurtosis	cv	rolling_vol_7	rolling_vol_30
-	Інше/невідомо	other	123	0	nan	5.262032520325205	5.08	0.7708234465939691	3.05	7.38	4.355000000000001	4.720000000000001	5.94	6.538	0.22578340144087045	0.04304271352944822	0.146487777036074	0.11183743324997715	0.10551139923428733
-	Вершки	cream	20	0	nan	54.33700000000001	37.2	30.05893896295988	23.34	101.94	23.397	32.349999999999994	93.945	99.907	0.6896399060295858	-1.3526822407178356	0.5531946732973825	0.8787289114990284	nan
-	Масло вершкове	butter	31	0	nan	317.1074193548387	289.0	87.12171256228238	198.30	598.8000000000000	229.5	259.0	338.97	498.0	1.6916176195835828	0.2747388022000028	0.3118541073650984	0.3571232254712644	nan
-	Молоко питне	milk	14	0	nan	73.88285714285713	70.56	36.36210610566391	36.66	120.0	37.205999999999996	39.0	101.88	120.0	0.11590575431406122	-2.126662695354071	0.49215890548676944	0.577001164010383	nan
-	Сир кисломолочний	cottage_cheese	11	0	nan	118.50636363636363	126.96	20.29961884998202	85.0	150.0	89.9	104.1	129.5	145.825	-0.1764338812511516	-0.8821627282689182	0.17129560157858975	0.26051494187992635	nan
-	Сир твердий	hard_cheese	46	0	nan	269.9910869565217	258.18	49.14078085701035	214.0	475.0	225.0	248.95499999999998	270.75	385.0	2.671068923901233	8.054536749966092	0.18200890040834491	0.1964883783704427	0.2298474616605193
-	Сметана	sour_cream	30	0	nan	105.84899999999999	100.08	22.27987438056582	57.0	149.0	79.84800000000001	95.685	112.5	146.1	0.5283846865444983	0.2164345365729874	0.2104873393283434	0.32759822957750356	nan
Івано-Франківська	Інше/невідомо	other	77	0	nan	6.13909090909091	6.22	0.8565355276039908	4.4	7.7	4.88	5.39	6.76	7.420000000000001	-0.06332253777988718	-1.0671678869465335	0.13952155788011103	0.13501142195235818	0.14075689337226976
Івано-Франківська	Вершки	cream	42	0	nan	35.28095238095237	34.32	5.000885682300008	27.24	46.86	28.695	31.275	38.64	43.943999999999996	0.4842887378520771	-0.27601247401112783	0.14174463399689594	0.18316725094280836	0.18678726420379502
Івано-Франківська	Масло вершкове	butter	70	0	nan	375.10585714285713	358.8	66.26840947376205	279.0	580.2	298.35	320.0	418.5	484.60499999999996	1.0069550722882605	0.9503953763013788	0.17666588834021876	0.23052262164322257	0.2094621469060926
Івано-Франківська	Молоко питне	milk	49	0	nan	47.52938775510204	43.26	24.07900056828246	28.8	144.0	30.54	36.0	48.0	95.99999999999996	3.306038059129264	10.9645867437682	0.5066128916356113	0.377530635079959	0.46771463323862644
Івано-Франківська	Сир кисломолочний	cottage_cheese	29	0	nan	123.81655172413794	117.0	20.315498003083764	86.04	154.8	94.6	111.0	145.0	151.992	0.028972545093300342	-1.1887327816900202	0.1640774009628898	0.2394727137499849	nan
Івано-Франківська	Сир твердий	hard_cheese	19	0	nan	299.4673684210526	291.0	40.40455376454838	239.0	366.0	239.576	276.0	325.5	361.30199999999996	0.30297193722009097	-0.9177907552914077	0.1349213905260602	0.1365686406707588	nan
Івано-Франківська	Сметана	sour_cream	17	0	nan	119.53176470588235	123.0	17.123022599447108	92.94	154.4	96.588	108.0	135.0	141.28	0.2375463977830526	-0.752653869866275	0.14325081405415288	0.17484610644693843	nan
Волинська	Інше/невідомо	other	122	0	nan	6.022704918032787	6.0	0.8384056463964787	4.350000000000001	8.120000000000001	4.702500000000001	5.41	6.7	7.294999999999999	-0.0006848267387692877	-0.7086693790373868	0.13920749195036597	0.1154776134492342	0.11044806372942716
Волинська	Вершки	cream	54	0	nan	38.88648148148148	38.269999999999996	6.906625246247507	26.58	61.1	29.04	35.425	41.837500000000006	51.699999999999999	1.1615188140328652	0.17760992800392376	0.23555216115623281	0.2121662092749512	0.2121662092749512
Волинська	Масло вершкове	butter	95	0	nan	393.6588421052632	388.0	67.05673117184827	220.0	575.0	270.0	349.65	447.18	489.09	0.027175912612710983	0.10689690456314738	0.1703422456186504	0.224615991305283	0.217872825283622
Волинська	Молоко питне	milk	37	0	nan	45.69108108108109	42.61	22.00839117546151	29.04	170.58	31.959999999999997	37.2	48.0	54.0	5.318039122651334	30.79895416409069	0.48167805739607106	0.37122172336602266	0.41882993287191755
Волинська	Сир кисломолочний	cottage_cheese	7	0	nan	103.64285714285714	95.0	31.49570643452094	84.0	174.0	84.87	88.45	97.8	151.67999999999995	2.4817734189143517	6.349439272263103	0.3038868987479622	nan	nan
Волинська	Сир твердий	hard_cheese	9	0	nan	278.2922222222222	296.22	29.208462719632343	215.05	300.0	230.23000000000002	265.2	299.0	299.976	-1.487151560011889	1.783268234978916	0.10495608711735309	0.16174524869261295	nan
Волинська	Сметана	sour_cream	16	0	nan	91.80625	82.9	23.64615454994744	70.0	159.0	71.5	75.0	102.3	130.875	1.7372893901484199	3.32382893399822	0.2575658470958942	0.3050119100803857	nan
Вінницька	Інше/невідомо	other	224	0	nan	5.9834375	5.92	0.9434738147535205	4.4	9.5	4.5915	5.2	6.705	7.5	0.34045959941789666	-0.27200527089185256	0.15768090077877817	0.1167961230941666	0.11985987941238893
Вінницька	Вершки	cream	67	0	nan	49.13283582089552	41.7	24.26128446011165	26.94	135.0	33.0	36.21	46.379999999999995	106.68599999999998	2.1705110471489095	3.6844562211058056	0.4937896226578816	0.544596819320073	0.5392978541691071
Вінницька	Масло вершкове	butter	155	0	nan	388.7090967741936	399.96	89.42348235548359	220.0	600.0	232.1	331.5	436.95000000000005	513.09	-0.306501915658805	-0.7166434778779531	0.2300524559306388	0.3349021388284887	0.34161777093102536
Вінницька	Молоко питне	milk	152	0	nan	53.66223684210526	48.0	25.673516263798142	31.92	156.0	36.275	42.0	48.51	125.95099999999995	2.632284436364918	5.7892922015111051	0.4784279928423298	0.3540578655620825	0.3427707942067689
Вінницька	Сир кисломолочний	cottage_cheese	41	0	nan	154.3029268292683	136.68	45.40121882966924	95.0	240.0	97.9	115.08	209.34	209.34	0.26665026209047643	-1.5337564043630258	0.29423433348029987	0.43088366161487685	0.42809494842050094
Вінницька	Сир твердий	hard_cheese	64	0	nan	304.541875	296.85	49.18356680892125	226.2	400.0	240.0	272.4	338.25	396.72	0.44176423899093265	0.7465605528262551	0.1615001773037657	0.23608461491535268	0.23499022676704917
Вінницька	Сметана	sour_cream	24	0	nan	130.06916666666666	132.42000000000002	18.23967770421146	86.2	168.0	105.045	119.25	138.81	163.54199999999997	-0.0022237530371951976	0.9158513313054573	0.14023060323710795	0.17171791622073312	nan
Дніпропетровська	Інше/невідомо	other	171	0	nan	5.713684210526315	5.7	0.927482928168138	3.95	8.4	4.234999999999999	4.9350000000000005	6.35	7.2	0.263344053032623	-0.5094998899364347	0.16232659943989153	0.18844437419561066	0.18109162198920836
Дніпропетровська	Вершки	cream	127	0	nan	48.01779527559056	41.4	28.67916471502275	27.9	189.0	29.748	34.41	48.5	87.83999999999999	3.6221218454883313	14.039415501910664	0.5972611726636596	0.5427932618805732	0.5595838062501391
Дніпропетровська	Масло вершкове	butter	214	0	nan	366.67004672897195	389.28	123.28209359216484	105.0	710.0	165.3	275.25	443.0	560.1	0.06663476002426949	-0.2895878463194821	0.3362207922134695	0.5032280327183776	0.5040122517570647
Дніпропетровська	Молоко питне	milk	90	0	nan	46.04255555555555	44.22	12.59138037261563	28.86	131.0	33.609	37.025	52.725	60.0	3.5268049565095935	22.314708212357457	0.27347266503099954	0.2996817103397616	0.3057811918831042
Дніпропетровська	Сир кисломолочний	cottage_cheese	53	0	nan	124.83433962264151	110.0	60.898508302731216	60.0	260.0	60.0	69.0	170.4	228.66	0.4752997250366984	-1.0791851007442443	0.48783458531378254	0.6835941796123562	0.6317646106187046
Дніпропетровська	Сир твердий	hard_cheese	56	0	nan	268.1533928571429	240.0	99.03465873166526	117.0	500.0	148.0	188.0	325.5	448.0	0.6285523716725998	-0.5475637308874752	0.3693209236566527	0.5663857212987943	0.6374315511203323
Дніпропетровська	Сметана	sour_cream	58	0	nan	120.485	127.02000000000001	39.29082028632044	65.52	210.0	71.75399999999999	77.7	149.25	180.0	0.13051527773069538	-1.2389449846515785	0.32610549268639616	0.5212954271666449	0.5143143159682037
Донецька	Інше/невідомо	other	6	0	nan	8.1	7.699999999999999	3.4578895297565535	4.9	13.05	4.950000000000001	5.150000000000001	10.1375	12.325000000000001	0.4076514311186301	-1.9347565898317454	0.42689994194525355	nan	nan
Донецька	Вершки	cream	2	0	nan	82.0	82.0	24.041630560342615	65.0	99.0	66.7	73.5	90.5	97.3	nan	nan	0.2931906165895441	nan	nan
Донецька	Масло вершкове	butter	10	0	nan	395.34	418.0	66.80131735227981	229.0	460.8	276.745	400.625	423.75	449.19	-2.0514243617057843	4.445271286055433	0.1689718150257495	0.20641447381018854	nan
Донецька	Молоко питне	milk	4	0	nan	43.262499999999996	42.0	6.433943192164508	36.85	52.2	37.6225	40.7125	44.55	50.67	1.1169056306323417	2.2107814008548896	0.14871871001824924	nan	nan
Донецька	Сир кисломолочний	cottage_cheese	1	0	nan	169.84	169.84	0.0	169.84	169.84	169.84	169.84	169.84	169.84	nan	nan	0.0	nan	nan

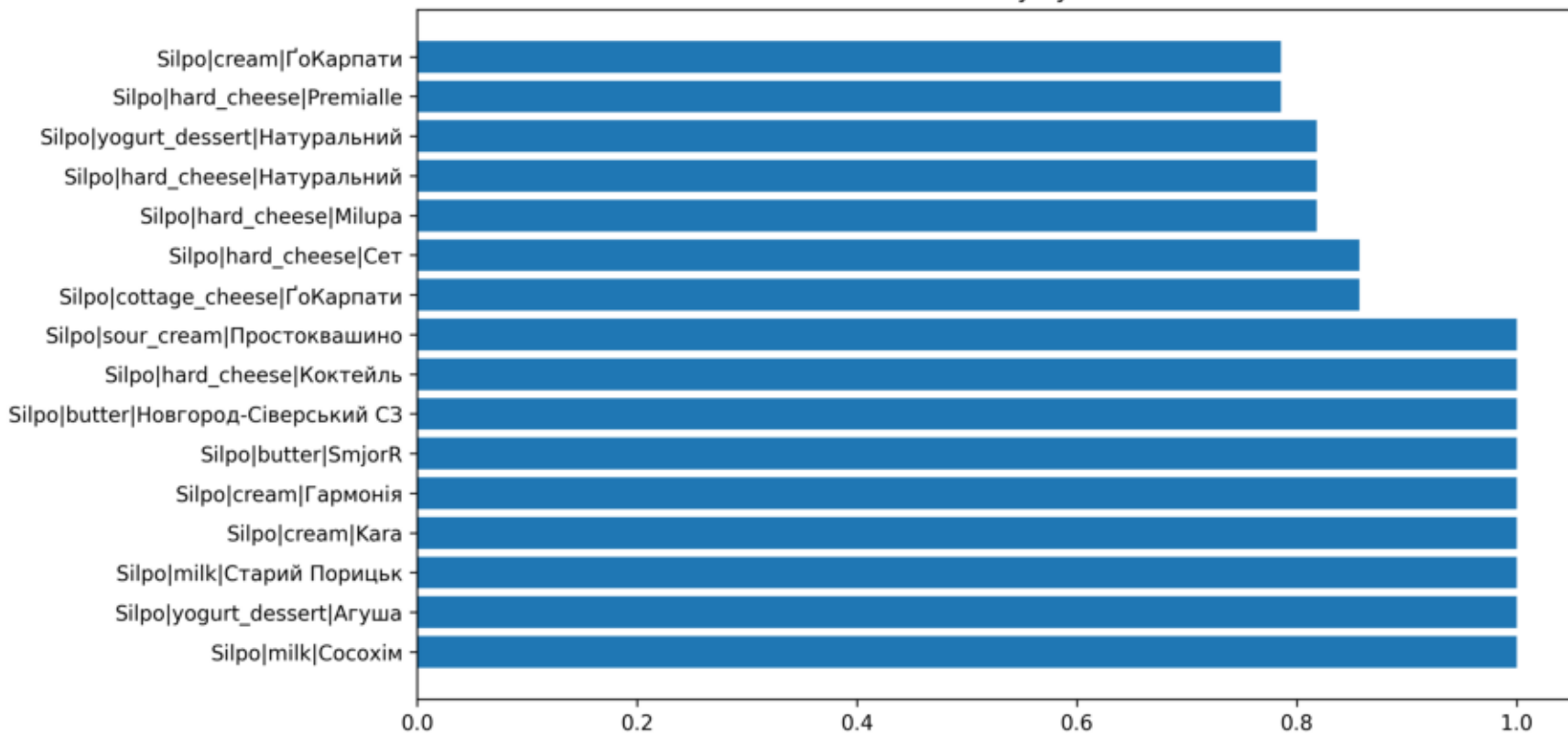
brand_hhi.png

Brand Concentration (HHI)



brand_promo_intensity.png

Promo Intensity by Brand



prozorro_region_median.png

ProZorro Median Unit Price by Region/Product

